LESSON 4 • TERM 1 • LEVEL 1

100 Square Explorer

Whole numbers: patterns in rows and columns

TODAY WE WILL

Explore the 100 square and find the patterns hiding in its rows and columns.



Get ready

0–4 MIN

- Your 100 square (the kit page)
- About 12 counters, all one kind
- Two or three colours
- Workbook, pencil and paper

Counters stay on the table, never near the mouth or ears.

4–10 MIN

Warm-up: count in tens

Show both hands for every ten

10	20	30	40	50	60	70	80	90	100

Now count back from 100. Whisper, and shout on 50!



Count along with me!

4–10 MIN

Fives and twos

One hand is five fingers. Count in fives:

5 10 15 20 25 30

Socks come in pairs. A pair is two. Count in twos:

2 4 6 8 10 12

4–10 MIN

What comes next?

10 20 30 ___

5 10 15 ___

2 4 6 ___

28 29 ___

38 39 ___

Say the next number. Show it on your fingers if you like.

Sneak-peek puzzles

10–14 MIN

Ideas we have not learnt yet. A guess is a great start. Nobody is marked.

PUZZLE 1: anyone can try

Count in 5s. Write the next two numbers.

5	10	15	20	___	___

PUZZLE 2: think

This is a piece of the 100 square. A counter is on 36.

Ravi moves it down one square.

Which number is it on now? How do you know?

21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40

Explore: read the square

14–28 MIN

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100

1. Put a finger on 1. Slide it along to 10.
2. Where does 11 sit? Slide your finger down.
3. Find 100. Which corner is it in?

Explore: cover the tens

14–28 MIN

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100

1. Count in tens: 10, 20, 30, ...
2. Put a counter on each number you say.
3. What do you see?

SAY IT

The counters make a ___.

Explore: fives and twos

14–28 MIN

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50

1. Clear your counters. Count in fives to 50.
2. What do you see?
3. Clear again. Count in twos to 20.
4. Which numbers have no counter?

Will 37 get a counter? Will 60?

14–28 MIN

Explore: move the counter

11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50

1. Counter on 34. Move it right 1.
2. Now down 1. Down 1 again.
3. Left 1. Then up 1.

WHAT DO YOU NOTICE?

Which way adds 1?

Which way adds 10?

28–38 MIN

Across: 1 more, 1 less

21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50





Right: 1 more. Left: 1 less.

At the end of a row, the next number starts the next row: 39, 40, 41.

28–38 MIN

Down: 10 more, up: 10 less

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100



Down: 10 more. Up: 10 less.

A row has ten squares. So one step down is ten more.

28–38 MIN

Your turn: write the number

Write on your paper. Timer: 1 minute

1

10 more than 36

2

10 less than 58

3

1 more than 29

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100

28–38 MIN

Check: write the number

1

10 more than 36

46

2

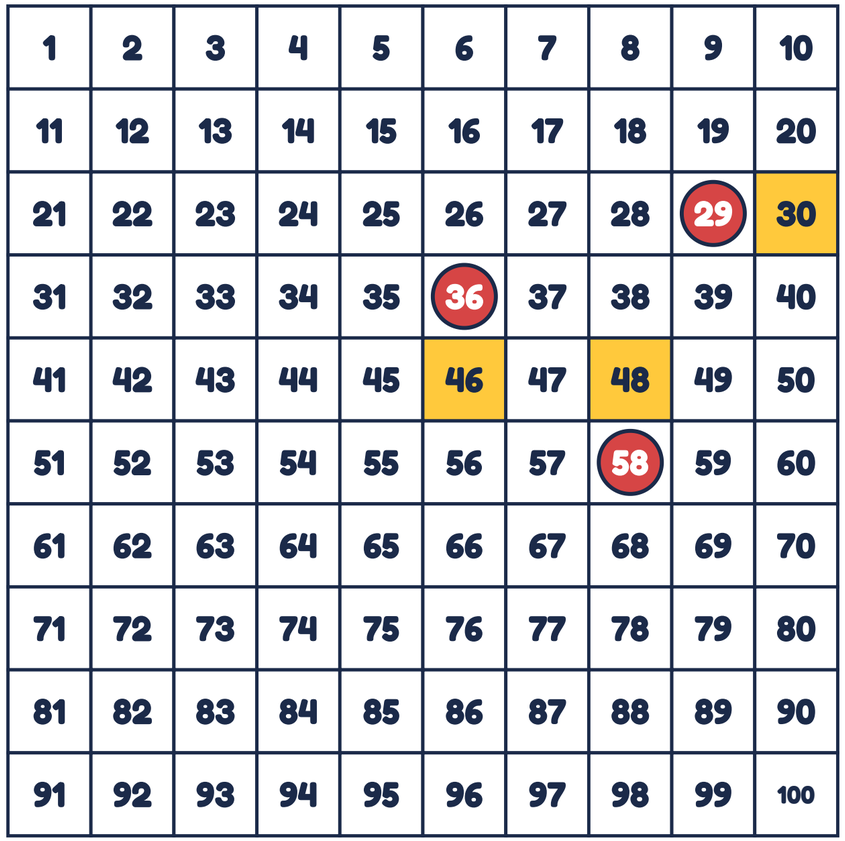
10 less than 58

48

3

1 more than 29

30



1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100

28–38 MIN

Stripes and last digits

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100

10s: one column

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100

5s: two columns

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100

2s: stripes

THE LAST DIGIT

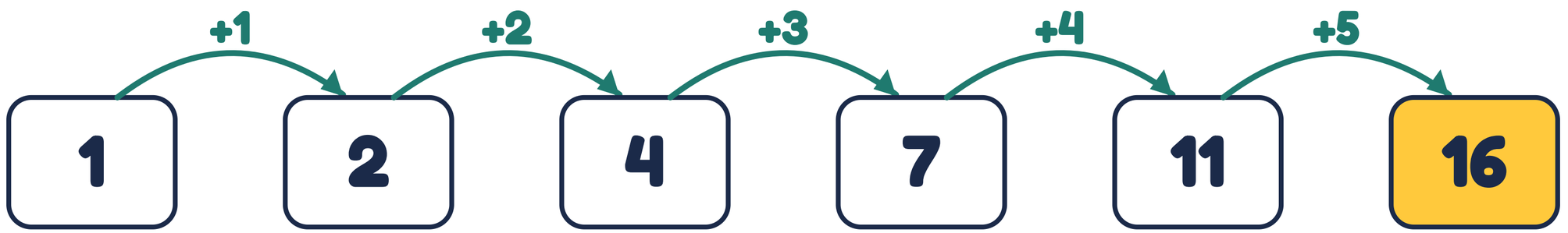
10s end in 0.

5s end in 0 or 5.

2s end in 0, 2, 4, 6 or 8.

28–38 MIN

Gaps can grow



The gap grows by 1 each time. So the next gap is 5.

28–38 MIN

Quick-fire: 10 more

23 47 **65** 90

Say 10 more. Think: one row down.

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100

38–42 MIN

Jump, clap, tap

A big jump for every ten. A clap for every five. A tiny tap for every two.

10 5 2

Sip some water.

42–70 MIN

Workbook page 1: Start here

Questions 1 and 2. Timer: 8 minutes

S1 Count in 10s. Colour 10, 20, 30, and keep going to 100. Then fill in the blanks.

S2 Count in 5s. Colour 5, 10, 15, and keep going to 100. Then fill in the blanks.

42–70 MIN

Check page 1

S1	They end in 0. They make a line going down.
S2	They end in 5 or 0. They make 2 lines going down.

42–70 MIN

Workbook page 2: Start here

Questions 3 to 6. Timer: 10 minutes

S3 Write the next two numbers in each pattern.

S4 Count in 2s. Colour 2, 4, 6, and keep going to 30. Write the last digits you see.

S5 This is a piece of the 100 square. Write the number in each empty square.

S6 Siti counts in 5s: 5, 10, 15, 20, 26, 30, 35. One number is wrong. Circle it. Write the number she should have said. How do you know?

Check page 2

42–70 MIN

S3	10, 12 | 30, 35 | 60, 70 | 41, 51
S4	2, 4, 6, 8, 0 (any order)
S5	a 34, b 43, c 55
S6	26 is wrong. She should have said 25. Counting in 5s ends in 0 or 5.

42–70 MIN

Choose your next page

Page	What it is
Think harder (page 3)	Our idea in a new setting
Go further (page 4)	Bigger puzzles
Investigation (page 5)	Counting in 3s
Check	Finish and check your own work

Everyone may choose. You can swap at any time.

70–76 MIN

Talk and explain

ALI SAYS

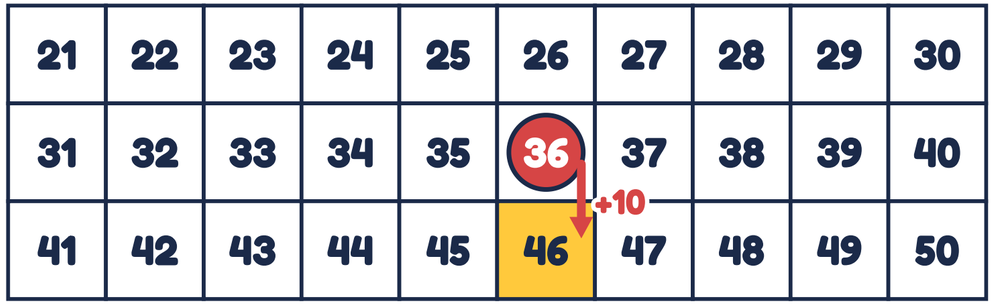
Counting in 5s makes one column on the square, just like counting in 10s.

Do you agree? Use the 100 square to explain.

I think ... because ...

Back to the sneak-peek

70–76 MIN



The counter is on 46. Down one row is 10 more.

76–82 MIN

Exit ticket

On your own. No looking back. Timer: 3 minutes

E1 Count in 5s. Write the next two numbers.

E2 A counter is on 26. It moves down 1, then right 1.

E3 Mei counts in 2s. She says 25. Is Mei right? Say why.

Check the exit ticket

76–82 MIN

E1	50, 55
E2	36, then 37
E3	No. 25 ends in 5. Counting in 2s ends in 0, 2, 4, 6 or 8.

Look-back

82–85 MIN

Three minutes. There are no right answers. You can pass. Nobody reads it out.

Something I did well	write or draw one thing
The hardest part was …	☐ finding my place on the square ☐ counting in 2s, 5s or 10s ☐ moving down a row ☐ the pattern where the gap grows (Think harder) ☐ something else
What helped me	☐ the 100 square or counters ☐ drawing it ☐ trying again ☐ I asked for help ☐ I was not stuck
Next time I will try	☐ drawing it ☐ using counters ☐ checking my answer ☐ asking a question
	☐ I am still stuck (my tutor will come and help) ☐ I would like to pass

Take your time. You can pass.



85–90 MIN

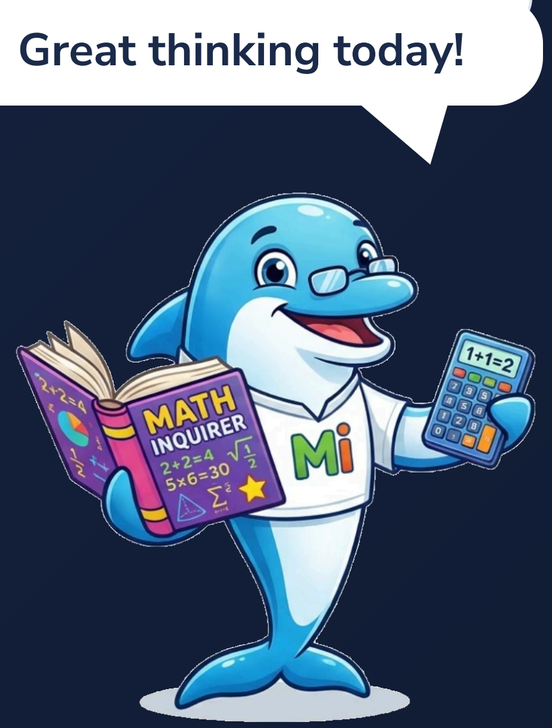
Well done, inquirers!

Show a grown-up one thing you found out today.

NEXT LESSON

Adding by making tens

You need a pencil and paper.



Go further: what if?

OPTIONAL

1	2	3	4	5
6	7	8	9	10
11	12	13	14	15
16	17	18	19	20

What if each row had only 5 numbers?

A counter is on 8. It moves down one row. Which number is it on?

What does going down add now?

OPTIONAL

If we have time: pick one

Activity	What we do
Is it always true?	Add 10 to a number. Does only the tens digit change?
Make a puzzle	Hide a number on the square and write clues for your grown-up
Teach it back	Show why counting in 10s makes a column

Optional. Everyone may join.